

Generalized Grade-of-Membership Estimation for High-dimensional Locally Dependent Data

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Abstract

This work focuses on the popular mixed membership models for multivariate categorical data widely used for analyzing survey responses and population genetics data. These grade of membership (GoM) models offer rich modeling power but present significant estimation challenges for high-dimensional polytomous data. Such data take the form of a three-way (quasi-)tensor, with many subjects responding to many items with varying numbers of categories. Popular estimation approaches such as Bayesian MCMC inference are not scalable to big data and lack theoretical guarantees in high dimensional settings. We introduce a novel and simple approach that flattens the three-way quasi-tensor into a “fat” matrix. We then perform singular value decomposition of this matrix to estimate parameters by exploiting the singular subspace geometry. Our fast spectral method can accommodate a broad range of data distributions with arbitrarily locally dependent noise, which we formalize as the generalized-GoM models. We establish finite-sample entrywise error bounds for the generalized-GoM model parameters. This is enabled by our sharp two-to-infinity singular subspace perturbation theory for locally dependent and flexibly distributed noise, a contribution of independent interest. Simulations and applications to data in political voting, population genetics, and single-cell genomics demonstrate our method’s superior performance.

Keywords: Grade of Membership model; Spectral method; Singular value decomposition; Local dependence; Two-to-infinity singular subspace perturbation.

1 Introduction

Multivariate categorical data are prevalent in various fields, including political surveys (Clinton et al., 2004), population genetics (Hartl et al., 1997), molecular biology (Becht et al., 2019), text analysis (Blei et al., 2003), among others. These data consist of many categorical

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variables collected across multiple individuals, and typically take the form of a three-way (quasi-)tensor, with many subjects responding to a large number of items with varying numbers of categories. Identifying interpretable latent patterns within such data is often of great interest. Among latent variable models for multivariate discrete data, the grade of membership (GoM) model (Woodbury et al., 1978) stands out as a popular and flexible tool that assumes the latent variables of each individual live in a probability simplex. The GoM model falls within the broad family of mixed membership (MM) models for individual-level mixtures (Airoldi et al., 2015). Unlike traditional population-level mixture models, such as the latent class model (Lazarsfeld, 1950), the GoM model allows for each individual to have partial memberships across multiple latent groups. In a GoM model with K extreme latent groups, each individual has K membership scores that are non-negative and sum to one, quantifying the extent to which the individual belongs to each group.

The flexibility of GoM models, however, brings significant challenges in estimation, particularly for high-dimensional polytomous data. Due to the complex hierarchical modeling nature of GoM models, most existing studies adopted the Bayesian approach with Markov Chain Monte Carlo (MCMC) algorithms for posterior inference (Erosheva, 2002; Erosheva et al., 2007; Bhattacharya and Dunson, 2012; Gormley and Murphy, 2009; Gu et al., 2023). Despite their popularity, these methods face critical challenges in high-dimensional settings. MCMC methods often suffer from high computational cost, making them hard to scale to big data. Furthermore, Bayesian or likelihood-based methods are usually tailored to a specific probabilistic model, limiting their applicability across different data distributions. Extending GoM model estimation methods developed for dichotomous data (Robitzsch and Robitzsch, 2017; Chen and Gu, 2024) to polytomous data also requires careful new development.

Another challenge in GoM models is local dependence among items in practice, where observed responses exhibit dependencies that are not fully explained by the latent variables. Local dependence has long been known as a practical issue when applying latent variable measurement models in social science (Chen and Thissen, 1997; Chen et al., 2018). For instance, in the American national election studies (ANES) dataset, blocks of items con-

cerning voting participation, presidential approval, economics, and other topics may display local dependence within each topic. In addition to social science, in genome-wide association studies (GWAS), linkage disequilibrium (LD) – where neighboring alleles are correlated – also represents a form of local dependence (Uffelmann et al., 2021). In single-cell data, gene co-expression that reflects dependencies between genes (Van Der Wijst et al., 2018) also constitutes local dependence. It is crucial to devise accurate estimation methods for locally dependent GoM models, especially in high-dimensional settings where such dependencies are often more pronounced.

To address the above challenges for GoM models, this paper introduces a fast spectral method based on singular value decomposition (SVD) for high-dimensional data with local dependence. We first observe that the data can be formulated as a three-way (quasi-)tensor, with many subjects responding to many items with varying numbers of categories. We employ a simple yet novel technique to flatten this quasi-tensor into a “fat” matrix and then perform matrix SVD to estimate parameters by exploiting the singular subspace geometry. We further introduce the generalized-GoM models, which encompass a wide range of data distributions with potentially locally dependent entries. We establish finite-sample entrywise error bounds of our method for generalized-GoM parameters. To achieve this, we develop sharp two-to-infinity singular subspace perturbation theory for arbitrary locally dependent data from flexible distributions, which is of independent interest. Extensive simulation studies demonstrate that our method accurately estimates the generalized GoM parameters with local dependence that either results from the flattening technique or is a consequence of the intrinsic data generation mechanism. The simulation results demonstrate the superior efficiency of the proposed method. We apply our method to a diverse range of real-world data in genetics, political science, and single-cell sequencing, and uncover interesting insights.

The rest of the paper is structured as follows. Section 2 introduces the model setup and explains the background and motivation behind the current work. Section 3 discusses model identifiability and presents an estimation algorithm for generalized-GoM models. Section 4 provides two-to-infinity singular subspace perturbation theory for locally dependence data

from flexible distributions and establishes entrywise error bounds for parameter estimation. Section 5 reports the results of simulation studies and Section 6 presents three real data applications. Section 7 concludes the paper. All technical proofs of the theoretical results and additional computation details are provided in the Supplementary Material.

2 Motivation and Model Setup

2.1 GoM Model for Multivariate Categorical Data

We first review the classical GoM model for multivariate categorical data, originally proposed in Woodbury et al. (1978). Consider categorical responses from N subjects to L items, and denote the response of subject i to item l by $\tilde{R}_{i,l} \in \{1, \dots, C_l\}$, for $i \in [N], l \in [L]$, where $C_l \geq 2$ denotes the number of categories for item l .

The GoM model uses latent mixed membership structures to model multivariate categorical data. A GoM model has two levels of modeling: the population level and the individual level. On the population level, there are K *extreme latent profiles*, each representing a prototypical response pattern. For each $k \in [K]$, the extreme profile k corresponds to an item parameter vector $\tilde{\boldsymbol{\theta}}_k = (\tilde{\boldsymbol{\theta}}_{1,k}^\top, \dots, \tilde{\boldsymbol{\theta}}_{L,k}^\top)^\top$, where $\tilde{\boldsymbol{\theta}}_{l,k} = (\tilde{\theta}_{l,k,1}, \dots, \tilde{\theta}_{l,k,C_l})^\top \in [0, 1]^{C_l}$, and

$$\tilde{\theta}_{l,k,c} = \mathbb{P}(\tilde{R}_{i,l} = c \mid \text{subject } i \text{ solely belongs to the } k\text{-th extreme latent profile}),$$

for $c \in [C_l], l \in [L]$. Note that $\sum_{c=1}^{C_l} \tilde{\theta}_{l,k,c} \equiv 1$ always holds for each $l \in [L]$ and $k \in [K]$. On the individual level, each subject i is associated with a vector of mixed membership scores $\boldsymbol{\pi}_i^* = (\pi_{i1}^*, \dots, \pi_{iK}^*)$, with $\pi_{ik}^* \geq 0$ and $\sum_{k=1}^K \pi_{ik}^* = 1$. Each membership score π_{ik}^* indicates the extent to which subject i partially belongs to extreme profile k . The item response probability for the i -th subject's response to the l -th item can be written as

$$\mathbb{P}(\tilde{R}_{i,l} = c) = \sum_{k=1}^K \pi_{ik}^* \tilde{\theta}_{l,k,c}, \text{ for } c \in [C_l]. \quad (1)$$

We collect all item parameters in matrix $\tilde{\Theta} = (\tilde{\vartheta}_1, \dots, \tilde{\vartheta}_K) \in [0, 1]^{(\sum_{l=1}^L C_l) \times K}$ and collect the mixed membership scores of all subjects in matrix $\Pi^* = (\pi_{ik}^*) \in \mathbb{R}^{N \times K}$.

There are two types of likelihood-based estimation methods for the GoM model, those that treat the membership scores as random effects and those that treat them as fixed effects (Erosheva, 2002). It is typically assumed that a subject's responses $\tilde{R}_{i,1}, \dots, \tilde{R}_{i,L}$ are conditionally independent given the membership scores π_i^* , which is called the local independence assumption. Under this assumption, if the membership scores π_i^* are considered to be i.i.d random vectors, e.g. π_i^* follows the Dirichlet distribution $D_{\alpha^*}(\cdot)$ parameterized with $\alpha^* \in \mathbb{R}_+^K$, then the marginal likelihood function of a GoM model is

$$L(\tilde{\Theta}, \alpha^* | \mathbf{R}) = \prod_{i=1}^N \int_{\Delta_{K-1}} \prod_{l=1}^L \prod_{c=1}^{C_l} \left(\sum_{k=1}^K \pi_{i,k}^* \tilde{\theta}_{l,k,c} \right)^{\mathbb{1}(\tilde{R}_{i,l}=c)} dD_{\alpha^*}(\pi_i). \quad (2)$$

The marginal maximum likelihood (MML) approach estimates the population parameters $(\tilde{\Theta}, \alpha^*)$ by maximizing the marginal likelihood function (2). However, the integrals over the membership scores are intractable, posing challenges for MML methods. Bayesian inference with MCMC is arguably the most widely used estimation approach for GoM models (Airolidi et al., 2015). Examples of Bayesian MCMC for GoM model and its variants include Erosheva (2002), Erosheva et al. (2004), Bhattacharya and Dunson (2012), Gu et al. (2023).

In addition to Bayesian MCMC methods, variational inference is another heuristic estimation approach (Barnard et al., 2003; Erosheva et al., 2007; Wang and Erosheva, 2015). On the other hand, when the membership scores are treated as fixed parameters, maximizing the joint maximum likelihood (JML) method can simultaneously estimate Π^* and $\tilde{\Theta}$ (Manton et al., 1994; Cooil and Varki, 2003). To our knowledge, there is a lack of readily available estimation tools that can handle polytomous data within the JML framework. The JML estimation function for GoM models in R package `sirt` (Robitzsch and Robitzsch, 2017) is only able to deal with dichotomous data. Likelihood-based methods generally scale poorly to large datasets, and the estimation errors are unknown in high dimensions. As an alternative, Zhao et al. (2018) proposed a moment estimation method that leverages the second

and third order moments to estimate both the item parameters and Dirichlet parameters in Dirichlet latent variable models with mixed data types.

2.2 Mixed Membership Models for Other Data Types

Mixed membership (MM) models have also been used to model data from the Binomial and Poisson distributions in some applications. For example, the number of minor allele of a single nucleotide polymorphism (SNP) is usually assumed to follow a Binomial distribution. Let $\Theta^* = (\theta_{j,k}^*)$ and $\theta_{j,k}^* \in (0, 1)$ represents the binomial parameter for the j -th item and the k -th extreme profile, then the genotype $R_{i,j}$ independently follows $\text{Binom}(2, \sum_{k=1}^K \pi_{i,k}^* \theta_{j,k}^*)$ for $i \in [N], j \in [J]$. The marginal likelihood function for a Binomial MM model is

$$L(\Theta^*, \alpha^* | \mathbf{R}) = \prod_{i=1}^N \int_{\Delta_{K-1}} \prod_{j=1}^J \binom{2}{R_{i,j}} \left(\sum_{k=1}^K \pi_{i,k}^* \theta_{j,k}^* \right)^{R_{i,j}} \left(1 - \sum_{k=1}^K \pi_{i,k}^* \theta_{j,k}^* \right)^{2-R_{i,j}} dD_{\alpha^*}(\pi_i)$$

when assuming $\pi_{i,k}^*$ follows distribution $D_{\alpha^*}(\cdot)$ parametrized by α^* . The influential MCMC algorithm STRUCTURE (Pritchard et al., 2000) imposes a Dirichlet prior on $\pi_{i,k}^*$ and estimates this MM model for Binomial distributed genotype data. As an example for count data, $R_{i,j}$ represents the expression count for the gene j in the cell i in single cell data analysis. In this case, $\theta_{j,k}^* \in \mathbb{R}_+$ denotes the Poisson rate and assume $R_{i,j} \stackrel{\text{ind.}}{\sim} \text{Poisson}(\pi_i^{*\top} \theta_j^*)$ for $i \in [N], j \in [J]$. The joint likelihood function for a Poisson MM model is given by

$$L(\Theta^*, \Pi^* | \mathbf{R}) = \prod_{i=1}^N \prod_{j=1}^J \frac{1}{R_{i,j}!} \left(\sum_{k=1}^K \pi_{i,k}^* \theta_{j,k}^* \right)^{R_{i,j}} \exp \left(- \sum_{k=1}^K \pi_{i,k}^* \theta_{j,k}^* \right). \quad (3)$$

To maximize (3), nonnegative matrix factorization (NMF) with multiplicative updates is a popular approach (Lee and Seung, 1999, 2000), where the estimates for Π^* and Θ^* are iteratively updated to approximate a local maximum of the joint likelihood. However, this approach suffers from initialization issues and is highly dependent on the Poisson distribution assumption. Similarly, using Bayesian MCMC for a Binomial or Poisson MM model necessitates algorithms specifically tailored to the noise distribution, and existing MCMC

algorithms designed for the MM model with Bernoulli distributed data cannot be directly applied to these cases.

2.3 Generalized-GoM Models and Assumptions

To facilitate statistical analysis of high-dimensional polytomous data, we introduce a simple but novel approach that flattens the three-way (quasi-)tensor into a “fat” binary matrix. Define $J := \sum_{l=0}^L C_l$ with $C_0 = 0$ and consider the binary matrix $\mathbf{R} = (R_{i,j}) \in \mathbb{R}^{N \times J}$ with

$$R_{i, \sum_{m=0}^{l-1} C_m + c} = \begin{cases} 1, & \text{if } \tilde{R}_{i,l} = c; \\ 0, & \text{if } \tilde{R}_{i,l} \neq c, \end{cases} \quad \text{for } c \in [C_l].$$

Similarly, we define the corresponding flattened item parameter matrix $\Theta^* = (\theta_{jk}^*) \in [0, 1]^{J \times K}$ with $\theta_{\sum_{m=1}^{l-1} C_m + c, k}^* = \tilde{\theta}_{l,k,c}^*$ for $l \in [L], k \in [K]$. The columns of $\mathbf{R}$ consist of L blocks, and within the l -th block we always have $\sum_{c=1}^{C_l} R_{i, \sum_{m=0}^{l-1} C_m + c} \equiv 1$. The size of the l th block is equal to the number of categories of the l th item. It is obvious that this flattening technique introduces local dependence within each block that does not stem from the latent variables. Figure 1 visualizes flattening a three-way (quasi-)tensor into a fat binary matrix, where the number of categories for the 4 items is 5, 3, 2, 3 respectively. As an illustration, item 2 can be asking “*which political party would do a better job handling illegal immigration?*” in the ANES data example, and the response categories are “Democrat”, “Republican”, and “no difference”. The cubes in the middle that correspond to item 2 represent the one-hot encoding of the three responses for all respondents. Similar flattened or concatenated structures have also been considered in the literature of the multilayer random graph models (Jones and Rubin-Delanchy, 2020; Gallagher et al., 2021).

To accommodate flexible local dependence structures and various data types, we introduce a class of *generalized-GoM* models that support general noise distributions, including

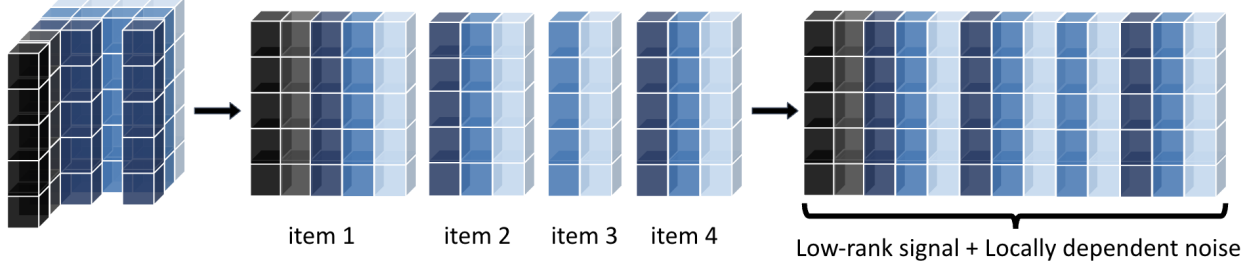


Figure 1: Visualization of the flattening technique. Cubes on the left represent the three-way quasi-tensor $\tilde{\mathbf{R}}$, and those on the right represent the flattened binary matrix $\mathbf{R}$.

those with sub-exponential tails. Consider the mean matrix and the error matrix

$$\mathbf{R}^* := \mathbb{E}[\mathbf{R}] = \underbrace{\mathbf{\Pi}^*}_{N \times K} \underbrace{\mathbf{\Theta}^{*\top}}_{K \times J}, \quad \mathbf{E} := \mathbf{R} - \mathbf{R}^*, \quad (4)$$

where $\mathbf{\Pi}^* = (\pi_{i,k}^*) \in [0, 1]^{N \times K}$ includes the membership scores, and $\mathbf{\Theta}^* = (\theta_{j,k}^*) \in \mathbb{R}_+^{J \times K}$ collects all the item parameters. Such expectation-noise decomposition is a commonly used in the topic modeling and network modeling literature, generally assuming independent entries in the noise matrix $\mathbf{E}$. In contrast, we impose the following Assumption 1 to characterize the general block dependence structure of the noise entries, along with their tail behavior.

Assumption 1 (Block Dependent Noise under Flexible Distributions). *Matrix $\mathbf{E}$ satisfies:*

- (a) *There exists a partition $S_1, S_2, \dots, S_L$ of $[J]$ with $|S_l| \leq M$ for each $l \in [L]$, such that the vectors $\{\mathbf{E}_{i,S_l}\}_{l=1}^L$ are mutually independent for different $i \in [N]$ and $l \in [L]$.*
- (b) *Either $\|E_{i,j}\|_\infty \leq B$ for all $i \in [N]$, $j \in [J]$, or there exists a random matrix $\mathbf{E}' = (E'_{i,j}) \in \mathbb{R}^{N \times J}$ obeying the same dependence structure in (a), such that for any $i \in [N]$, $j \in [J]$, it holds that $\|E'_{i,j}\|_\infty \leq B$, $\mathbb{E}[E'_{i,j}] = 0$, $\|\text{Cov}(\mathbf{E}'_{i,:})\| \lesssim \|\text{Cov}(\mathbf{E}_{i,:})\|$, and $\Pr(E_{i,j} = E'_{i,j}) \geq 1 - O(d^{-22})$.*

The generalized-GoM models satisfying Assumption 1 are suitable for handling highly asymmetric and sparse data, and are applicable across diverse fields, including population genetics, single cell analysis, social science surveys, among others. The flexible local dependence in Assumption 1(a) effectively captures the dependence in the flattened data from multivari-

ate polytomous responses and accommodates other more complex dependence structures. The entries in each block S_l can be arbitrarily dependent on each other. They can either consist of only those flattened response categories of a single item and/or concatenated responses across different intrinsically locally dependent items. Related local dependence considerations are explored in [Lei \(2019\)](#); [Modell et al. \(2024\)](#). In contrast, [Agterberg et al. \(2022\)](#) introduces a linear-transformed entrywise-independent noise assumption, relaxing the entrywise independence assumption in a different direction, but their assumption cannot capture the dependence structure of flattened polytomous data in our motivating setting.

Notably, the high-probability boundedness Assumption 1(b) encompasses a wide range of data types, covering all sub-exponential distributions and substantially extending beyond the flattened polytomous data case in the traditional GoM model. For example, the mixed membership models for Binomial and Poisson data mentioned in Section 2.2 both belong to the generalized-GoM modeling framework. Specifically, one can write the data arising from these models as the sum of a low-rank signal part $\mathbf{\Pi}^* \mathbf{\Theta}^{*\top}$ and a mean-zero Binomial/Poisson noise part $\mathbf{E}$, where parameters $\mathbf{\Theta}^*$ have their respective range.

Consider the singular value decomposition (SVD) of the low-rank signal matrix $\mathbf{R}^*$:

$$\mathbf{R}^* = \mathbf{U}^* \mathbf{\Lambda}^* \mathbf{V}^{*\top}, \quad (5)$$

where $\mathbf{\Lambda}^* = \text{diag}(\sigma_1^*, \dots, \sigma_K^*)$ with $\sigma_1^* \geq \dots \geq \sigma_K^* > 0$ collects the K singular values. Orthonormal matrices $\mathbf{U}_{N \times K}^*$, $\mathbf{V}_{J \times K}^*$ respectively consist of the left and right singular vectors. Similarly, we consider the top- K SVD for the observed matrix $\mathbf{R} \approx \mathbf{U} \mathbf{\Lambda} \mathbf{V}^\top$ where $\mathbf{\Lambda} = \text{diag}(\sigma_1, \dots, \sigma_K)$ with $\sigma_1 \geq \dots, \sigma_K \geq 0$, $\mathbf{U} \in \mathbb{R}^{N \times K}$, $\mathbf{V} \in \mathbb{R}^{J \times K}$, and $\mathbf{U}^\top \mathbf{U} = \mathbf{V}^\top \mathbf{V} = \mathbf{I}_K$.

Write the ratio of J and N as $r := J/N$. Let $[B] = \{1, 2, \dots, B\}$ for any positive integer B . For a matrix $\mathbf{A}$, let $\lambda_k(\mathbf{A})$ denote the k -th largest eigenvalue and $\sigma_k(\mathbf{A})$ denote the k -th largest singular value; define its condition number $\kappa(\mathbf{A}) := \sigma_1(\mathbf{A})/\sigma_{\min}(\mathbf{A})$ with $\sigma_{\min}(\mathbf{A})$ being the minimum non-zero singular value. Specifically, define $\kappa^* := \sigma_1^*/\sigma_K^*$ as the condition number of $\mathbf{R}^*$. The $\ell_{2,\infty}$ norm of $\mathbf{A}$ is $\|\mathbf{A}\|_{2,\infty} := \max_i \|\mathbf{A}_{i,:}\|_2$, which is the maximum row

ℓ_2 norm of $\mathbf{A}$. Let $\mathbf{A}_{\mathbf{S},:}$ denote the submatrix consisting of the rows indexed by $\mathbf{S}$ and similarly $\mathbf{A}_{:, \mathbf{S}}$ for columns. Define the matrix incoherence degrees, which are standard in matrix perturbation theory: $\mu_1 := (N/K) \|\mathbf{U}^*\|_{2,\infty}^2$, and $\mu_2 := (J/K) \|\mathbf{V}^*\|_{2,\infty}^2$.

Finally, we define the following notations similar to those in [Chen et al. \(2021\)](#): (a) $f(n) = O(g(n))$ or $f(n) \lesssim g(n)$ means there is a universal constant $c > 0$ such that $|f(n)| \leq c|g(n)|$ for all n large enough. And $\tilde{O}(\cdot)$ differs from $O(\cdot)$ with up to poly-logarithmic factors; (b) $f(n) \asymp g(n)$ indicates that there exists universal constants $c_1, c_2 > 0$ such that $c_1|g(n)| \leq |f(n)| \leq c_2|g(n)|$ for all n large enough; (c) $f(n) = o(g(n))$ denotes that $f(n)/g(n) \xrightarrow{n \rightarrow \infty} 0$; (d) $f(n) \gg g(n)$ (resp. $f(n) \ll g(n)$) is used when there exists universal constant $c > 0$ large (resp. small) enough such that $|f(n)| \geq c|g(n)|$ (resp. $|f(n)| \leq c|g(n)|$).

3 A Spectral Method for Estimation

Model identifiability is crucial in statistical analysis, especially for latent variable modeling, as it is a prerequisite for reproducible and interpretable statistical inference. Conventionally, a statistical model is said to be identifiable if there is a one-on-one mapping between the population parameters and the marginal likelihood function. However, it is more than challenging to establish identifiability for GoM model because of the complicated integration and summation in (2). To tackle this problem, [Chen and Gu \(2024\)](#) proposes another notion of identifiability - expectation identifiability. With this notion, a GoM model with parameter set $(\mathbf{\Pi}^*, \mathbf{\Theta}^*)$ is said to be identifiable, if for any other valid parameter set $(\tilde{\mathbf{\Pi}}, \tilde{\mathbf{\Theta}})$, $\tilde{\mathbf{\Pi}}\tilde{\mathbf{\Theta}}^\top = \mathbf{\Pi}^*\mathbf{\Theta}^{*\top}$ holds if and only if $(\mathbf{\Pi}^*, \mathbf{\Theta}^*)$ and $(\tilde{\mathbf{\Pi}}, \tilde{\mathbf{\Theta}})$ are identical up to a permutation of the K extreme profiles. To establish expectation identifiability for generalized-GoM models, we define subject i to be a pure subject for extreme profile k if the only positive entry of $\boldsymbol{\pi}_i^*$ is located at index k . In the setting discussed in [Chen and Gu \(2024\)](#), all the entries in $\mathbf{E}$ are assumed to be independent. However, since identifiability is discussed upon the expectation matrix, the same identifiability results still hold when the noises are block-wisely independent. Assumption 2 below is sufficient for a generalized-GoM model to be identifiable, as

shown in Lemma 1.

Assumption 2 (Pure Subjects). Π^* satisfies that every extreme latent profile has at least one pure subject. In addition, $\text{rank}(\Theta^*) = K$.

The pure subject condition in Assumption 2 is analogous to the anchor word assumption in topic modeling (Ke and Wang, 2024) and the pure node assumption in mixed membership stochastic block model (MMSB) in network analysis (Jin et al., 2024). Under Assumption 2, denote $\mathbf{S} = (S_1, \dots, S_K)$ as the index vector of one set of K pure subjects such that $\Pi_{\mathbf{S},:}^* = \mathbf{I}_K$. Lemma 1 gives the model parameter expressions in terms of $\mathbf{S}$ and the population SVD.

Algorithm 1 Polytomous-GoM Estimation

Input: Polytomous response matrix $\tilde{\mathbf{R}} \in \mathbb{R}^{N \times L}$, number of extreme profiles K

Output: Estimated $\hat{\mathbf{S}} \in \mathbb{R}^K$, $\hat{\Theta}^{(\text{post})} \in \mathbb{R}^{J \times K}$, $\hat{\Pi}^{(\text{post})} \in \mathbb{R}^{N \times K}$

- 1: Obtain the flattened binary matrix $\mathbf{R} \in \mathbb{R}^{N \times J}$ from $\tilde{\mathbf{R}}$
 - 2: Get the top K singular value decomposition of $\mathbf{R}$ as $\mathbf{U}\mathbf{\Lambda}\mathbf{V}^\top$
 - 3: $\mathbf{Y} = \mathbf{U}$
 - 4: **for** $k \in \{1, \dots, K\}$ **do**
 - 5: $\hat{S}_k = \text{argmax}(\{\|\mathbf{Y}_{i,:}\|_2 : i \in [N] \setminus \hat{\mathbf{P}}\})$
 - 6: $\mathbf{u} = \mathbf{Y}_{\hat{S}_k,:} / \|\mathbf{Y}_{\hat{S}_k,:}\|_2$
 - 7: $\mathbf{Y} = \mathbf{Y}(\mathbf{I}_K - \mathbf{u}\mathbf{u}^\top)$
 - 8: **end for**
 - 9: $\hat{\Pi} = \mathbf{U}(\mathbf{U}_{\hat{\mathbf{S}},:})^{-1}$
 - 10: $\hat{\Pi}^{(\text{post})} = \text{diag}(\hat{\Pi}_+ \mathbf{1}_K)^{-1} \hat{\Pi}_+$
 - 11: $\hat{\Theta} = \mathbf{V}\mathbf{\Lambda}\mathbf{U}_{\hat{\mathbf{S}},:}^\top$
 - 12: Let $\hat{\Theta}^{(\text{post})} = (\hat{\theta}_{j,k}^{(\text{post})})$ with $\hat{\theta}_{j,k}^{(\text{post})} = \begin{cases} \hat{\theta}_{j,k} & \text{if } 0 \leq \hat{\theta}_{j,k} \leq 1 \\ 0 & \text{if } \hat{\theta}_{j,k} < 0 \\ 1 & \text{if } \hat{\theta}_{j,k} > 1 \end{cases}$
 - 13: **for** $l \in \{1, \dots, L\}$ **do**
 - 14: **for** $k \in \{1, \dots, K\}$ **do**
 - 15: $\theta^b = \hat{\Theta}_{(\sum_{t=1}^{l-1} C_t + 1):(\sum_{t=1}^l C_t),k}^{(\text{post})}$
 - 16: $\hat{\Theta}_{(\sum_{t=1}^{l-1} C_t + 1):\sum_{t=1}^l C_t,k}^{(\text{post})} = \theta^b / \mathbf{1}_K^\top \theta^b$
 - 17: **end for**
 - 18: **end for**
-

Lemma 1. *Under Assumption 2, the parameters have the closed-form expressions*

$$\mathbf{\Pi}^* = \mathbf{U}^* \mathbf{U}_{\mathbf{S},:}^{*-1}, \quad \mathbf{\Theta}^* = \mathbf{V}^* \mathbf{\Lambda}^* \mathbf{U}_{\mathbf{S},:}^{*\top}. \quad (6)$$

The first equation in (6) implies that the i -th row of left singular matrix $\mathbf{U}^*$ can be written as a convex combination of $\mathbf{U}_{S_1}^*, \dots, \mathbf{U}_{S_K}^*$, with the weights being the membership scores π_i^* . This geometric property motivates us to use a vertex hunting algorithm called the successive projection algorithm (SPA, [Gillis and Vavasis, 2013](#)) to find the indices $\mathbf{S}$ of the pure subjects. SPA successively finds the vertices with the largest norm in a sequentially defined orthogonal subspace. After estimating the indices $\hat{\mathbf{S}} = (\hat{S}_1, \dots, \hat{S}_K)$, we can estimate the parameters using (6). Algorithm 1 summarizes the proposed procedure for the polytomous-GoM (i.e., traditional GoM) model. Our estimation algorithm for Poisson-GoM is given in Section S.3.2 in the Supplementary Material. Algorithm 1 can be easily generalized to accommodate generalized-GoM models with other data types by modifying the post-processing on the item parameter estimate $\hat{\mathbf{\Theta}}$ (lines 12-17) to the appropriate range. The post-processing steps truncate the estimates to valid values and the estimation error can only gets smaller.

Similar spectral methods for MM models are well-studied for network models and topic models. The geometric properties in the spectral domain have been exploited for the MM network model using eigenvalue decomposition of the symmetric adjacency matrix ([Jin et al., 2024](#); [Mao et al., 2021](#); [Jin et al., 2024](#); [Ke and Jin, 2023](#); [Jin et al., 2021](#); [Zhang et al., 2020](#)). Also, SVD-based methods are used to estimate the topic models ([Ke and Wang, 2024](#); [Wu et al., 2023](#)). Additionally, some works ([Fan et al., 2022](#); [Du and Tang, 2023](#)) focus on the testing problem of the membership scores, instead of the parameter estimation problem. Compared to these existing works, we are motivated by asymmetric mixed membership models, especially the polytomous-GoM model for high-dimensional categorical responses. Data arising from this model cannot be directly written as a signal matrix plus an *entrywise-independent* noise matrix, in contrast to the usual network model and topic model with independent multinomial noise. For our motivating models, Bayesian methods

and likelihood-based methods are the predominant paradigms for parameter estimation. In contrast, our spectral method offers a unifying new scalable estimation approach that comes with strong theoretical guarantees under local dependence; see Section 4.

4 Uniform Estimation Error Bounds for Parameters

To establish entrywise estimation error bounds for $\hat{\boldsymbol{\Pi}}$ and $\hat{\boldsymbol{\Theta}}$, it is necessary to develop fine-grained perturbation bounds on the empirical singular matrices $\mathbf{U}$ and $\mathbf{V}$ according to Lemma 1. In Section 4.1, we establish a novel $\ell_{2,\infty}$ perturbation theory for the left and right singular subspaces for the truncated SVD under locally dependent and flexibly distributed noise. In Section 4.2, we present uniform entrywise convergence rates for parameters using the new singular subspace perturbation theory. Finally, in Section 4.3, we discuss an application to generalized-GoM models with Dirichlet priors.

4.1 $\ell_{2,\infty}$ Singular Subspace Perturbation under Locally Dependent and General-distribution Noise

We parse the impact of local dependence by measuring the covariances:

$$\sigma^2 := \max_{i \in [N], j \in [J]} \text{Var}(E_{i,j}), \quad \tilde{\sigma}^2 := \max_{i \in [N], l \in [L]} \|\text{Cov}(\mathbf{E}_{i,S_l})\|.$$

One can verify that $\sigma^2 \leq \tilde{\sigma}^2 \leq M\sigma^2$. In the special case of independent entries in $\mathbf{E}$, $\tilde{\sigma}^2 = \sigma^2$; in the another special case of exactly equal entries within each block, $\tilde{\sigma}^2 = M\sigma^2$.

Assumption 3. *We assume that $K \ll \min\{N, L\}$, $ML \asymp J$, and*

$$\sigma_K(\boldsymbol{\Pi}^*)\sigma_K(\boldsymbol{\Theta}^*) \gtrsim \sigma\kappa^*(M\sqrt{N} + \sqrt{J}), \quad (7)$$

$$MB \log^2(N \vee J) \max \left\{ \sqrt{\frac{\mu_1 K}{N}}, \sqrt{\frac{\mu_2 K}{J}} \right\} \lesssim \sigma. \quad (8)$$

Assumption 3 assumes a low-rank signal matrix and balanced block size in the noise

matrix. In addition, (7) assumes a lower bound on signal strength, and (8) assumes that the incoherence degrees cannot be too large. Theorem 1 below gives our novel $\ell_{2,\infty}$ singular subspace perturbation bounds. These results apply to general low-rank matrices with locally dependent noise from flexible distributions (see Assumption 1), and are of independent interest to other latent variable models beyond the mixed membership models.

Theorem 1 (Two-to-infinity Singular Subspace Perturbation Under Local Dependence).

Assume that Assumptions 1, 3 hold. Then with probability at least $1 - O((N \vee J)^{-10})$,

$$\begin{aligned} & \|\mathbf{U}\mathbf{U}^\top\mathbf{U}^* - \mathbf{U}^*\|_{2,\infty} \\ & \lesssim \frac{\tilde{\sigma}\sqrt{N\log(N \vee J)}}{\sigma_K^*} \sqrt{\frac{\mu_1 K}{N}} + \frac{\kappa^* \sigma^2 J}{\sigma_K^{*2}} \sqrt{\frac{\mu_1 K}{N}} + \frac{\sigma MB \log(N \vee J) \sqrt{MN+J}}{\sigma_K^{*2}} \sqrt{\frac{\mu_2 K}{J}} =: \xi_1, \end{aligned} \quad (9)$$

$$\begin{aligned} & \|\mathbf{V}\mathbf{V}^\top\mathbf{V}^* - \mathbf{V}^*\|_{2,\infty} \\ & \lesssim \frac{\sigma\sqrt{J\log(N \vee J)}}{\sigma_K^*} \sqrt{\frac{\mu_2 K}{J}} + \frac{\kappa^* \sigma^2 MN}{\sigma_K^{*2}} \sqrt{\frac{\mu_2 K}{J}} + \frac{\sigma B \log(N \vee J) \sqrt{MN+J}}{\sigma_K^{*2}} \sqrt{\frac{\mu_1 K}{N}} =: \xi_2, \end{aligned} \quad (10)$$

$$\begin{aligned} & \|\mathbf{U}\mathbf{\Lambda}\mathbf{V}^\top - \mathbf{U}^*\mathbf{\Lambda}^*\mathbf{V}^{*\top}\|_\infty \\ & \lesssim \kappa^* K \sigma \log(N \vee J) \sqrt{\mu_1 \mu_2} \sqrt{\frac{M}{J} + \frac{1}{N}} + MB \log(N \vee J) \left(\frac{\mu_2 K}{J} + \frac{\mu_1 K}{MN} \right) =: \xi_3. \end{aligned} \quad (11)$$

Remark 1. By imposing additional assumptions, we can derive simpler expressions for the upper bounds presented in Theorem 1. First, provided that $MB/\sigma \lesssim \log(N \vee J)$ and $\mu_2/\mu_1 \lesssim J/(\log(N \vee J))^3$, then the third term in (9) is dominated by the first term and

$$\|\mathbf{U}\mathbf{U}^\top\mathbf{U}^* - \mathbf{U}^*\|_{2,\infty} \lesssim \left(\frac{\tilde{\sigma}\sqrt{N\log(N \vee J)}}{\sigma_K^*} + \frac{\kappa^* \sigma^2 J}{\sigma_K^{*2}} \right) \sqrt{\frac{\mu_1 K}{N}}. \quad (12)$$

Similarly, if $B/\sigma \lesssim \log(N \vee J)$ and $\mu_1/\mu_2 \lesssim N/(\log(N \vee J))^3$ hold, (10) is simplified to

$$\|\mathbf{V}\mathbf{V}^\top\mathbf{V}^* - \mathbf{V}^*\|_{2,\infty} \lesssim \left(\frac{\sigma\sqrt{J\log(N \vee J)}}{\sigma_K^*} + \frac{\kappa^* \sigma^2 MN}{\sigma_K^{*2}} \right) \sqrt{\frac{\mu_2 K}{J}}. \quad (13)$$

Lastly, if $\sqrt{MB} \lesssim \kappa^* \sigma \sqrt{\frac{\mu_1}{\mu_2} \wedge \frac{\mu_2}{\mu_1}} \sqrt{\frac{M^2 N^2}{J} \wedge \frac{J^2}{MN}}$ holds, the entrywise bound (11) becomes

$$\left\| \mathbf{U} \mathbf{\Lambda} \mathbf{V}^\top - \mathbf{U}^* \mathbf{\Lambda}^* \mathbf{V}^{*\top} \right\|_\infty \lesssim \kappa^* K \sigma \sqrt{\mu_1 \mu_2} \log(N \vee J) \sqrt{\frac{M}{J} + \frac{1}{N}}. \quad (14)$$

Remark 2. We give an interpretation of (13) in Remark 1, and similar interpretation holds for (12). We employ the expansion formula in (Xia, 2021, Theorem 1):

$$\begin{aligned} & \mathbf{V} \mathbf{V}^\top \mathbf{V}^* - \mathbf{V}^* \\ &= (\mathbf{I}_J - \mathbf{V}^* \mathbf{V}^{*\top}) \underbrace{(\mathbf{E}^\top \mathbf{R}^* + \mathbf{E}^\top \mathbf{E} + \mathbf{R}^{*\top} \mathbf{E})}_{\mathbf{R}^\top \mathbf{R} - \mathbf{R}^{*\top} \mathbf{R}^*} \mathbf{V}^* \mathbf{\Lambda}^{*-2} + \text{higher order terms} \\ &= \mathbf{E}^\top \mathbf{R}^* \mathbf{V}^* \mathbf{\Lambda}^{*-2} + \mathbf{E}^\top \mathbf{E} \mathbf{V}^* \mathbf{\Lambda}^{*-2} + \text{negligible terms} + \text{higher order terms}. \end{aligned} \quad (15)$$

The two terms of the upper bound in (13) correspond to the decomposition above as follows:

$$\begin{aligned} \left\| \mathbf{E}^\top \mathbf{R}^* \mathbf{V}^* \mathbf{\Lambda}^{*-2} \right\|_{2,\infty} &= \left\| \mathbf{E}^\top \mathbf{U}^* \mathbf{\Lambda}^{*-1} \right\|_{2,\infty} \lesssim \frac{\sigma \sqrt{J \log(N \vee J)}}{\sigma_K^*} \sqrt{\frac{\mu_2 K}{J}}, \\ \left\| \mathbf{E}^\top \mathbf{E} \mathbf{V}^* \mathbf{\Lambda}^{*-2} \right\|_{2,\infty} &\lesssim \frac{\sigma \sqrt{K \log(N \vee J)}}{\sigma_K^*} + \frac{\sigma^2 M N}{\sigma_K^{*2}} \sqrt{\frac{\mu_2 K}{J}} \\ &\lesssim \frac{\sigma \sqrt{J \log(N \vee J)}}{\sigma_K^*} \sqrt{\frac{\mu_2 K}{J}} + \frac{\kappa^* \sigma^2 M N}{\sigma_K^{*2}} \sqrt{\frac{\mu_2 K}{J}}, \end{aligned}$$

hold with probability at least $1 - O((N \vee J)^{-10})$. The first line follows from inequality (S.8) in Lemma S.3 and the second line holds by Lemma S.7 in the Supplementary Material.

Allowing for arbitrarily flexible locally dependent noise is a key distinguishing feature of our bounds. In recent years, there has been significant progress in $\ell_{2,\infty}$ eigenspace perturbation theory for random matrices (Fan et al., 2018; Cape et al., 2019; Mao et al., 2021; Chen et al., 2021; Cai et al., 2021; Abbe et al., 2022; Yan et al., 2024). However, in this literature, most studies consider independent entries in the noise matrix, with only a few exceptions including Lei (2019), Modell et al. (2024), and Agterberg et al. (2022). To facilitate the interpretation, we delve further into the simplified forms presented in Remark 1. On the one

hand, our upper bound (12) for the left singular subspace $\mathbf{U}$ involves $\tilde{\sigma}$ as a critical quantity to measure the noise level, and provides a sharp characterization of the SVD-based method's performance (in light of a similar decomposition to (15) for $\mathbf{U}$). On the other hand, it is inevitable to involve the block size M in our leave-one-block-out argument for deriving the bound for the right singular subspace $\mathbf{V}$. Actually, we highlight that the dependence on M in our bounds (13) and (14) is *sharp*, by considering the following special case. Imagine the scenario where each block size is exactly M , and the entries $R_{i,j}$, $j \in S_l$ in each block S_l are duplicates of a single variable $\check{R}_{i,l} \sim \mathcal{N}(0, \sigma^2)$ for each $i \in [N]$. Invoking (15), the fluctuation $\mathbf{V}_{j,:} \mathbf{V}^\top \mathbf{V}^* - \mathbf{V}_{j,:}^*$ is partially attributed to the component $\mathbf{E}_{:,j}^\top \mathbf{E}_{:,S_l} \mathbf{V}_{S_l,:}^* \mathbf{\Lambda}^{*-2}$, given $j \in S_l$. Applying elementary concentration inequalities (e.g. Hoeffding's inequality for sub-Gaussian entries), we find that $(\mathbf{E}^\top \mathbf{E})_{S_l, S_l}$ is concentrated around $\sum_{i \in [N]} \text{Var}(\check{R}_{i,l}) \mathbf{1}_M \mathbf{1}_M^\top$. This leads to the result that $\max_{j \in [J]} \|\mathbf{E}_{:,j}^\top \mathbf{E}_{:,S_l} \mathbf{V}_{S_l,:}^* \mathbf{\Lambda}^{*-2}\|_2$ is lower bounded by $\frac{\sigma^2 M N}{\sigma_1^{*2}} \|\mathbf{V}^*\|_{2,\infty}$ with high probability. This confirms that the dependence of (13) on M is sharp and cannot be improved. Moreover, under the same setting, the minimax lower bound for the entry-wise error, as informed by Koltchinskii et al. (2011), is $\sigma \sqrt{K} \sqrt{M/J + 1/N}$. Provided that $\mu_1, \mu_2, \kappa^*, K = O(1)$, our upper bound (14) is optimal up to a logarithmic term.

We further clarify how our bounds distinguish from previous results. In the literature, Lei (2019) and Modell et al. (2024) studied eigenspace perturbation to accommodate certain dependence structures. To adapt these results from symmetric random matrices to asymmetric matrices, a common approach is to embed the asymmetric matrix as two blocks within a larger symmetric matrix, a technique known as the “symmetric dilation” trick. However, this symmetrization technique often falls short of delivering a sharp dependence on the ratio J/N and the incoherence degrees μ_1 and μ_2 , because the corresponding singular subspace perturbation bounds resulting from symmetrization for asymmetric random matrices typically involve $J \vee N$ and $\mu_1 \vee \mu_2$. However, distinguishing N from J and distinguishing μ_1 from μ_2 become particularly important when encountering unbalanced observation matrices ($N = o(J)$ or $J = o(N)$) with unbalanced incoherence degrees ($\mu_1 \ll \mu_2$ or $\mu_1 \gg \mu_2$). To see the importance of distinguishing μ_1 between μ_2 , we note that $\mu_1 \leq \kappa^2(\mathbf{\Pi}^*)/K$ when

Assumption 2 holds by Lemmas 1 and S.7, thus $\mu_1 = O(1)$ if $\kappa(\mathbf{\Pi}^*) = O(\sqrt{K})$. Section 4.3 will show that $\kappa(\mathbf{\Pi}^*) = O(\sqrt{K})$ holds with high probability if the membership scores are generated from a Dirichlet distribution, a common assumption in Bayesian inference for GoM models. By contrast, μ_2 often exhibits larger values due to the prevalence of sparse patterns and heterogeneity among the items. This distinction is supported by numerous real-world datasets, including the single cell data in Section 6 where the empirical value of μ_1 is around 2.9 while for μ_2 it is 55.7. We address this issue by establishing perturbation bounds for the left and right singular vectors *separately*. Our technique is conceptually similar to that of the arXiv version of Yan et al. (2024), with accommodation to the case with local dependence.

Lastly, we remark on the application of matrix concentration universality in our proof to overcome potential dependence issues. Specifically, given an N -by- J noise matrix $\mathbf{E}$ with block dependence in Assumption 1, we employ Corollary 2.15 in Brailovskaya and van Handel (2024) to derive that $\|\mathbf{E}\| \lesssim \tilde{\sigma}\sqrt{N} + \sigma\sqrt{J}$ with high probability. To compare this with the preceding approaches, consider the bound provided by the matrix Bernstein inequality, which often yields $\|\mathbf{E}\| \lesssim (\tilde{\sigma}\sqrt{N} + \sigma\sqrt{J})\sqrt{\log(N \vee J)}$ with high probability. This result is $\sqrt{\log(N \vee J)}$ -looser than our universality-based concentration. Moreover, this elimination of the $\sqrt{\log(N \vee J)}$ factor also extends to the bounds on $\|\mathbf{E}\mathbf{V}^*\|$ and $\|\mathbf{U}^{*\top}\mathbf{E}\|$.

4.2 Uniform Consistency of Parameter Estimation

We establish the error bounds for the estimators $\hat{\mathbf{\Pi}} = \mathbf{U}\mathbf{U}_{\hat{\mathbf{S}},:}^{-1}$, $\hat{\mathbf{\Theta}} = \mathbf{V}\mathbf{\Lambda}\mathbf{U}_{\hat{\mathbf{S}},:}^{\top}$ in Theorem 2.

Theorem 2 (Uniform Consistency of Parameter Estimation). *Suppose Assumptions 1, 2, 3 hold, and additional assume that $\sqrt{K}\kappa^2(\mathbf{\Pi}^*)\sigma_1(\mathbf{\Pi}^*)\xi_1 \ll 1$. Then there exists a permutation matrix $\mathbf{P}$ such that with probability at least $1 - O((N \vee J)^{-10})$,*

$$\left\| \hat{\mathbf{\Pi}} - \mathbf{\Pi}^*\mathbf{P} \right\|_{2,\infty} = O(\kappa^2(\mathbf{\Pi}^*)\sigma_1(\mathbf{\Pi}^*)\xi_1), \quad (16)$$

$$\left\| \hat{\mathbf{\Theta}} - \mathbf{\Theta}^*\mathbf{P} \right\|_{\infty} = O(\kappa^2(\mathbf{\Pi}^*)\xi_3). \quad (17)$$

The $\ell_{2,\infty}$ bound of $\hat{\mathbf{\Pi}}$ primarily depends on the left singular subspace perturbation, as the

estimate derives from $\widehat{\mathbf{U}}$. The additional term $\kappa^2(\mathbf{\Pi}^*)\sigma_1(\mathbf{\Pi}^*)$ arises from the vertex hunting step. Similarly, the ℓ_∞ error bound of $\widehat{\mathbf{\Theta}}$ involves the upper bound of $\|\mathbf{R}^* - \mathbf{U}\mathbf{\Lambda}\mathbf{V}^\top\|_\infty$, which is reasonable given that $\widehat{\mathbf{\Theta}}^\top$ is a submatrix of $\mathbf{U}\mathbf{\Lambda}\mathbf{V}^\top$, and $\kappa^2(\mathbf{\Pi}^*)$ again stems from estimating $\widehat{\mathbf{S}}$ via vertex hunting.

Corollary 1. *Suppose the assumptions in Theorem 2 hold and $\max\{\kappa(\mathbf{\Pi}^*), \kappa(\mathbf{\Theta}^*)\} = O(1)$. Then there exists a permutation matrix $\mathbf{P}$ such that with probability at least $1 - O((N \vee J)^{-10})$,*

$$\begin{aligned} & \left\| \widehat{\mathbf{\Pi}} - \mathbf{\Pi}^* \mathbf{P} \right\|_{2,\infty} \\ & \lesssim \frac{\tilde{\sigma} \sqrt{\mu_1 K \log(N \vee J)}}{\sigma_K(\mathbf{\Theta}^*)} + \frac{\sigma^2 \sqrt{JK}}{\sigma_K(\mathbf{\Pi}^*) \sigma_K^2(\mathbf{\Theta}^*)} \left(\sqrt{\frac{J}{N}} \sqrt{\mu_1} + \sqrt{1 + \frac{MN}{J}} \frac{MB \log(N \vee J)}{\sigma \sqrt{J}} \sqrt{\mu_2} \right), \end{aligned} \quad (18)$$

$$\begin{aligned} & \left\| \widehat{\mathbf{\Theta}} - \mathbf{\Theta}^* \mathbf{P} \right\|_\infty \\ & \lesssim K \sigma \log(N \vee J) \sqrt{\mu_1 \mu_2} \sqrt{\frac{M}{J} + \frac{1}{N}} + MB \log(N \vee J) \left(\frac{\mu_1 K}{MN} + \frac{\mu_2 K}{J} \right). \end{aligned} \quad (19)$$

To see the rates of convergence with Theorem 2, consider the case where $\sigma_K^2(\mathbf{\Pi}^*) \gtrsim N/K$, $\sigma_K^2(\mathbf{\Theta}^*) \gtrsim J/(MK)$. Under the mild assumptions for (12) and (14) in Remark 1, $\widehat{\pi}_i$ converges to π_i^* with a rate of $\tilde{O}\left(\sigma K^2 M \sqrt{\mu_1} \left(\frac{1}{\sqrt{J}} \vee \frac{\sigma K}{N}\right)\right)$, and $\widehat{\theta}_j$ converges to θ_j^* with a rate of $\tilde{O}\left(\sigma K^2 \sqrt{\mu_1 \mu_2} \left(\sqrt{\frac{M}{J}} \vee \frac{1}{\sqrt{N}}\right)\right)$.

4.3 Application to Generalized GoM Models with Dirichlet Prior

Dirichlet prior on the membership scores is commonly adopted in Bayesian inference for GoM models (Erosheva, 2002; Erosheva et al., 2004; Gu et al., 2023). To illustrate the theoretical results, we discuss a Dirichlet prior application on the GoM model for polytomous data estimated using the flattening procedure. Assume that π_i^* is independently sampled from a Dirichlet distribution with parameters $\alpha = (\alpha_1, \dots, \alpha_K)$. For simplicity, we assume $C_l = M$ for all $l \in [L]$. Similarly, we assume that for each $l \in [L], k \in [K]$, vector $(\theta_{M(l-1)+c,k}^*)_{c=1}^M$ independently follows a Dirichlet distribution with parameters $\beta = (\beta_1, \dots, \beta_M)$.

Assumption 4. Assume that $\max_k \alpha_k \leq C_1 \min_k \alpha_k$ for some $C_1 > 0$, $\max_c \beta_c \leq C_2 \min_c \beta_c$ for some $C_2 > 0$, $\max_k \alpha_k = O(1)$, $\max_c \beta_c = O(1)$. In addition, $N \gtrsim M^2 K^5 \log^4(N \vee J)$, $J \gtrsim M^4 K^6 \log^4(N \vee J)$ and

$$\nu \leq \frac{\sqrt{N/\log N}}{6\sqrt{3}(1 + \alpha_0)}, \quad c_1 \geq 4K\sqrt{15 \log L/L}. \quad (20)$$

Note that (20) is similar to Assumption 3.3 in Mao et al. (2021).

Corollary 2. Let $\pi_i^* \stackrel{i.i.d}{\sim} \text{Dirichlet}(\boldsymbol{\alpha})$ and $(\theta_{M(l-1)+c,k}^*)_{c=1}^M \stackrel{i.i.d}{\sim} \text{Dirichlet}(\boldsymbol{\beta})$. Suppose Assumption 4 holds. Then with probability at least $1 - O(K(N \wedge L)^{-3})$, one has

$$\begin{aligned} \|\hat{\boldsymbol{\Pi}} - \boldsymbol{\Pi}^* \mathbf{P}\|_{2,\infty} &\lesssim \frac{M^{3/2} K^{5/2} \sqrt{\log(N \vee J)}}{\sqrt{J}} + \frac{\sqrt{M^2 K^{9/2}}}{N}, \\ \|\hat{\boldsymbol{\Theta}} - \boldsymbol{\Theta}^* \mathbf{P}\|_{\infty} &\lesssim \frac{M^{5/2} K^{9/2} \log(N \vee J)}{\sqrt{J}} + \frac{M^2 K^{9/2} \log(N \vee J)}{\sqrt{N}}. \end{aligned}$$

5 Simulation Studies

We conduct three simulation studies to assess the efficiency and accuracy of the proposed method. Section 5.1 presents results for the GoM model for multivariate polytomous data, where the item responses are polytomous and local dependence is introduced by the flattening technique. Section 5.2 presents results for the GoM model for multivariate binary data, where the binary item responses are designed to have intrinsic blockwise dependent entries. In the third study in Section 5.3, we consider the generalized-GoM model for Poisson data and compare our method with the popular NMF algorithm (Lee and Seung, 2000). For all three studies, we consider $K = 3$ and generate 100 independent replications for each simulation setting. Within each setting and each replication, the rows of $\boldsymbol{\Pi}$ are independently simulated from the $\text{Dirichlet}(\mathbf{1}_K)$ distribution, and the first K rows of $\boldsymbol{\Pi}$ are set to the identity matrix $\mathbf{I}_K$ to satisfy Assumption 2. For both the simulation studies and real-data applications, we employ a pruning step on $\mathbf{U}$ before SPA in Algorithm 1 to de-noise (Mao et al., 2021), where we set the tuning parameters as $r = 10$, $q = 0.4$, $e = 0.2$ as suggested in Chen and Gu (2024).

5.1 Local Dependence from Flattening Polytomous Responses

We first consider local dependence introduced by the flattening technique for polytomous data, with the number of response categories set to $C_l \equiv C = 3$ for all $l \in [L]$. Consider sample size $N \in \{200, 1000, 2000, 3000, 4000, 5000\}$, and $J = N/5$. For each $j \in [J]$ and $k \in [K]$, $\Theta_{(C(j-1)+1):(Cj),k}^*$ is independently simulated from $\text{Dirichlet}(0.2 \cdot \mathbf{1}_C)$. For each pair (i, j) , the non-zero index in $\mathbf{R}_{i,(C(j-1)+1):(Cj)}$ is independently generated from the categorical distribution with the probability vector being $\Theta_{(C(j-1)+1):(Cj),:}^* \Pi_{i,:}^\top$.

Figure 2 displays the boxplots of the 2-to-infinity estimation error for $\hat{\Pi}$ and the maximum absolute estimation error for $\hat{\Theta}$ obtained from the proposed method. We observe that the error decreases and approaches 0 as N, J grow for both estimates, which verifies our uniform consistency results for parameter estimation.

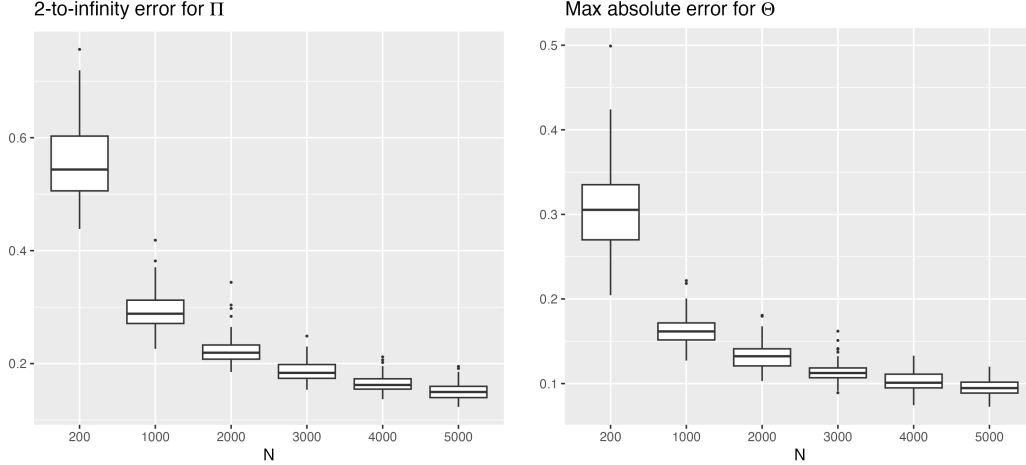


Figure 2: Simulation study I, estimation error for the proposed method using the flattening technique. The left figure presents the 2-to-infinity error for $\hat{\Pi}$ and the right presents the maximum absolute error for $\hat{\Theta}$.

We compare our method with the Gibbs sampling method and the variational inference method for the polytomous GoM model. The parameters in the Gibbs sampler are initialized with their generating distributions. The derivation of each updating step in the Gibbs sampler can be found in Section S.3.2 in the Supplementary Material. We set the first 5000 samples as burn-in and take the average of 2000 samples after the burn-in phase as the Gibbs sampling estimates. We employ the variational inference (VI) algorithm proposed

	$N = 200$	$N = 1000$	$N = 2000$
Proposed	0.07	2.25	12.84
VI	6.63	85.14	264.40
Gibbs	709.44	16458.22	65464.17

Table 1: Average computation time in seconds for each method and sample size for polytomous responses. The flattening technique is used in the proposed method. VI stands for the variational inference algorithm implemented in the `mixedMem` R package.

Θ	$N = 200$	$N = 1000$	$N = 2000$	Π	$N = 200$	$N = 1000$	$N = 2000$
Proposed	0.057	0.030	0.024	Proposed	0.098	0.047	0.035
VI	0.252	0.265	0.272	VI	0.204	0.198	0.198
Gibbs	0.072	0.032	0.023	Gibbs	0.080	0.038	0.028

Table 2: Average mean absolute error for Θ and Π for each method and sample size for polytomous responses. The flattening technique is used in the proposed method. VI stands for the variational inference algorithm implemented in the `mixedMem` R package.

in Wang and Erosheva (2015), implemented in the `mixedMemModel` function in R package `mixedMem`. As Gibbs sampling is not scalable to large datasets and would take a very long time to compute when N, J are large, we only consider $N \in \{200, 1000, 2000\}$ to compare the three methods’ performances. Table 1 presents the average computation time in seconds across the 100 replications. When $N = 2000$, our proposed method is about 10,000 times faster compared to Gibbs sampling and 30 times faster compared to VI. Table 2 summarizes the average 2-to-infinity error of $\hat{\Pi}$ and the max absolute error of $\hat{\Theta}$. In comparison to Gibbs sampling, the proposed method delivers comparable results in accuracy. We also find that the VI method in this case does not yield consistent results, and it is very sensitive to initialization. In summary, our proposed method is efficient in computation, especially for large-scale datasets, and is comparably accurate with popular existing methods.

5.2 Intrinsic Local Dependence across Different Items

In this section, we generate binary responses with a more general intrinsic local dependence structure. Now $\Theta^* \in [0, 1]^{J \times K}$ is no longer a flattened parameter matrix, but rather contains the item response probabilities as the Bernoulli parameters $\theta_{jk}^* \stackrel{i.i.d}{\sim} \text{Beta}(0.2, 0.2)$. The membership scores π_i^* are generated with the same mechanism as stated before. To generate

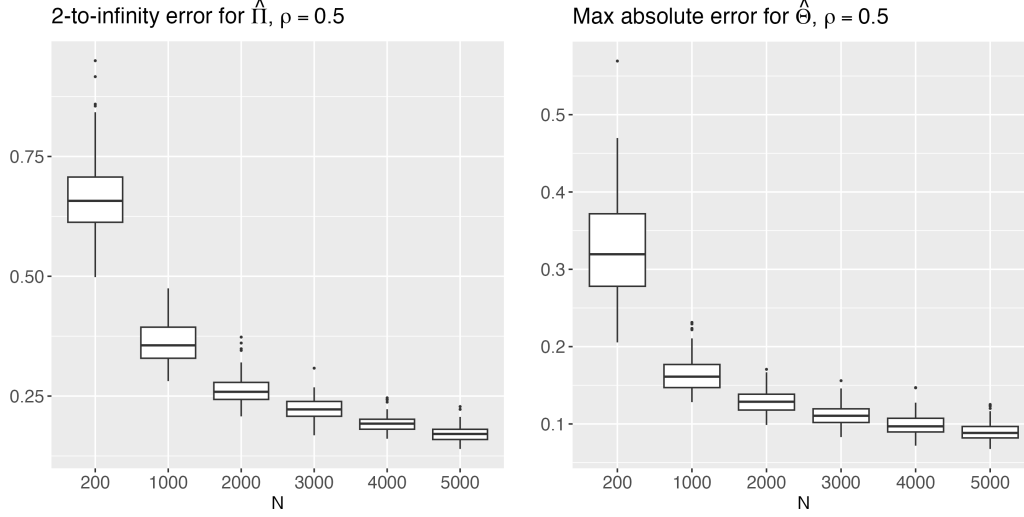


Figure 3: Simulation results of estimation error for the proposed method with block-wise dependent data. The left figure represents the 2-to-infinity error for $\hat{\Pi}$ and the right represents the maximum absolute error for $\hat{\Theta}$.

the locally dependent responses, we first generate latent variables $\boldsymbol{\eta}_i \in \mathbb{R}^J$ independently and identically from the Gaussian distribution $N(\mathbf{0}, \boldsymbol{\Sigma})$, with $\boldsymbol{\Sigma}$ being blockwise diagonal with J/M copies of a $M \times M$ block covariance matrix $\boldsymbol{\Sigma}_0$. Here $\boldsymbol{\Sigma}_0 \in \mathbb{R}^{M \times M}$ has an autoregressive structure with parameter ρ , that is, $(\boldsymbol{\Sigma}_0)_{ij} = \rho^{|i-j|}$. We then define a threshold $d_{ij} = \Phi^{-1}(\boldsymbol{\Pi}_{i,:}^\top \boldsymbol{\Theta}_{j,:})$ where Φ is the cumulative distribution function for the standard Normal distribution. Finally, each binary response is defined as $R_{i,j} = \mathbb{1}(\eta_{ij} < d_{ij})$. From this construction, we have $\mathbb{E}[R_{i,j}] = \Pr(R_{i,j} = 1) = \boldsymbol{\Pi}_{i,:}^\top \boldsymbol{\Theta}_{j,:}$, and the elements in $\mathbf{R}_{i,:}$ are blockwisely dependent with a block size M . Specifically, we set $\rho = 0.5$ and $M = 10$, which induces a relatively strong level of dependence. We still set $N \in \{200, 1000, 2000, 3000, 4000, 5000\}$ and let $J = N/5$. Figure 3 presents the error boxplots in this scenario. We can see that the estimates from our proposed method are consistent as N and J grow.

5.3 Generalized-GoM for Poisson Data

In this experiment, we consider simulated data with independent Poisson entries, and compare the proposed method with the multiplicative update NMF algorithm proposed in Lee and Seung (2000). The Poisson rate parameters $\theta_{j,k}^*$ are independently sampled from

$\stackrel{i.i.d}{\sim}$ Gamma(1, 2). To see the effect of the ratio between N and J , we consider both $N = 5J$ with $N \in \{200, 1000, 2000, 3000\}$ and $J = 5N$ with $N \in \{200, 400, 600, 800\}$. We use the R package **NMF** to implement the NMF algorithm (Gaujoux and Seoighe, 2010), where most of the built-in algorithms have been optimized in C++. We compare the two methods in terms of computation time and the scaled Frobenius error $\frac{1}{\sqrt{NJ}} \left\| \mathbf{R}^* - \hat{\mathbf{\Pi}} \hat{\mathbf{\Theta}}^\top \right\|_F$, demonstrated in Figure 4. We can see that the proposed method yields slightly more accurate estimates across different simulation settings. The computation time of the two methods is comparable when $N = 5J$, while the proposed method is much more efficient in the high-dimensional setting with $J = 5N$. It is worth noting that the matrix decomposition estimate from NMF requires post-processing to locate each row of $\hat{\mathbf{\Pi}}$ on a simplex and that the NMF algorithm itself does not give consistent estimates of parameters $\mathbf{\Pi}$ and $\mathbf{\Theta}$.

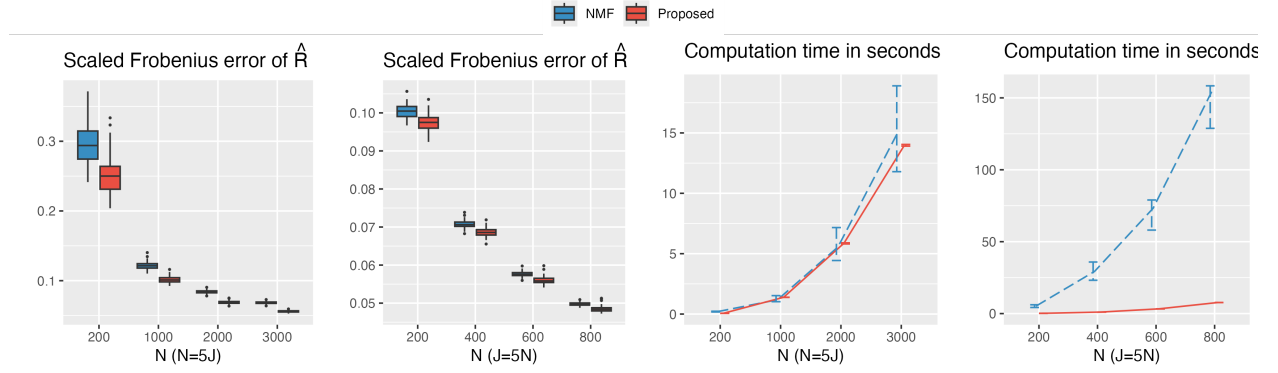


Figure 4: Comparing the estimation error and computation time for the proposed method and NMF with $N = 5J$ and $J = 5N$.

6 Real-data Applications

In this section, we apply the proposed method to three real-world datasets arising from diverse applications, including population genetics data ($N \ll J$, binomial data), national election survey data ($N > J$, polytomous data), and single-cell sequencing data ($N \ll J$, nonnegative count data). These datasets exhibit various different local dependence behaviors, either resulting from the flattening technique, or from intrinsic dependence, or both.

6.1 Population Genetics: HapMap3 Genotype Data

We consider a genotyping dataset from the HapMap3 project (Consortium et al., 2010, publicly available at <https://www.broadinstitute.org/medical-and-population-genetics/hapmap-3>). A subset of 4 sub-populations are considered, including the African ancestry in Southwest USA (ASW), Utah residents with Northern and Western European ancestry (CEU), Mexican ancestry in Los Angeles, California (MEX), and Yoruba in Ibadan, Nigeria (YRI). We follow the conventional data preprocessing procedure to remove SNPs with minor allele frequency smaller than 5% and SNPs with missing data (Kranzler et al., 2019). The pre-processed dataset consists of $N = 467$ individuals and $J = 274,128$ SNPs.

The observed genotype R_{ij} for individual i at locus j takes value in $\{0, 1, 2\}$, which is the count of mutations relative to the reference allele. The genotype of each SNP is often assumed to follow a binomial distribution (Jiang et al., 2024). In this case, we have $\mathbf{R}^* = \mathbb{E}[\mathbf{R}] = 2\mathbf{\Pi}^*\mathbf{\Theta}^{*\top}$ with $\mathbf{\Theta}^*$ consisting of the binomial parameters. Linkage disequilibrium (LD) is an important and well-studied phenomena for SNP genotype data, which refers to correlations among neighbouring alleles (Reich et al., 2001). Therefore, we consider a binomial model with local dependence for this dataset, and directly employ our proposed method on $\mathbf{R}/2$ to estimate $\mathbf{\Pi}^*$ and $\mathbf{\Theta}^*$. The left panel in Figure 5 is the ternary plot based on the estimated membership scores with our proposed method. Our method is able to reveal that the ASW sample is a mixture of the YRI and CEU samples, and the MEX sample is a mixture between the CEU sample and a third ancestral population. These observations align with the established biological understanding of human population genetics (Jakobsson et al., 2008; Li et al., 2008; Alexander et al., 2009). Furthermore, the left panel in Figure 8 confirms the existence of local dependence among the SNPs in the data, as Figure S.1 in the Supplementary Material demonstrates that the covariance matrix of the estimated error matrix $\hat{\mathbf{E}}$ is a good estimate of the true error covariance matrix.

We compare the proposed method with a highly influential method called STRUCTURE (Pritchard et al., 2000; Falush et al., 2003). STRUCTURE is a method for multilocus genotype data to infer population structure and assign admixed proportions (membership

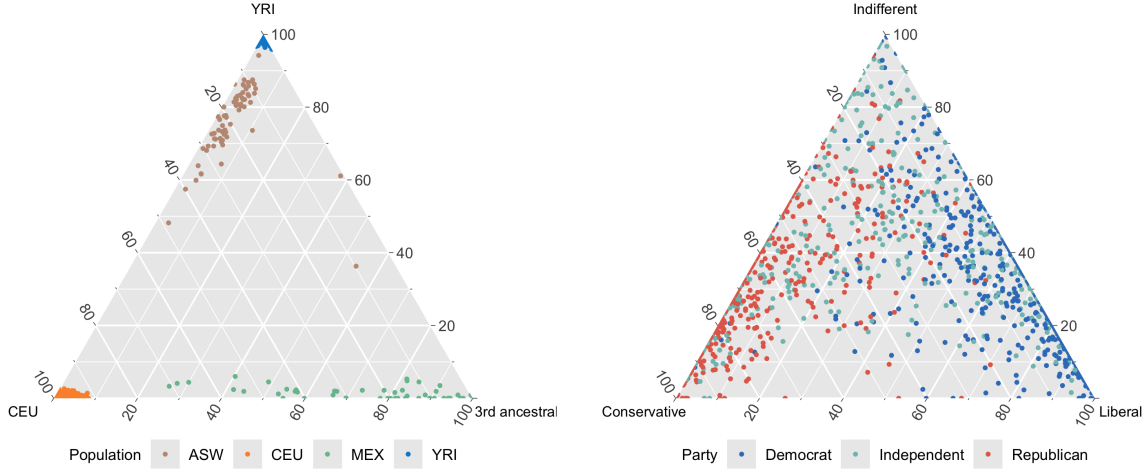


Figure 5: Ternary plots of the estimated membership scores for the HapMap3 data (left) and ANES data (right). For the Hapmap3 dataset, ASW denotes African ancestry in Southwest USA; CEU denotes Utah residents with Northern and Western European ancestry; MEX denotes Mexican ancestry in Los Angeles, California; YRI denotes Yoruba in Ibadan, Nigeria. For the ANES data, each dot is colored by the subject’s self-reported party information.

scores) to each individual. Essentially, STRUCTURE assumes a GoM model, where the admixture proportions are assumed to be sampled from $\text{Dirichlet}(\alpha, \dots, \alpha)$, and the item parameters from $\text{Dirichlet}(\lambda_1, \dots, \lambda_{C_j})$. The hyper-parameter α is assumed to have a uniform hyper-prior $\text{Unif}[0, 10]$. STRUCTURE uses an MCMC algorithm to estimate the model parameters and employs Metropolis-Hasting to update α at each iteration. Figure 6 displays the comparisons of our proposed method with STRUCTURE in terms of the membership score estimates. The figures are generated with the **Distruct** software (Rosenberg, 2004). Each vertical slice represents an individual, and each color represents one extreme latent profile. The lengths of the three color bars for each individual i are proportional to the estimated membership score $\hat{\pi}_{i,k}$ for $k = 1, 2, 3$. We can see that both methods capture rather similar admixture structures in the sample. However, STRUCTURE took over 44 hours and 22 minutes to run with default settings, yet the proposed method takes only about 9 seconds. Our proposed method is able to generate similar results with a huge computation advantage. On another note, if we treat the $\{0, 1, 2\}$ responses as polytomous data and use the flattening technique before estimating parameters, it turns out that it gives very similar estimation results of $\hat{\Pi}$ to those given by the binomial model reported above.

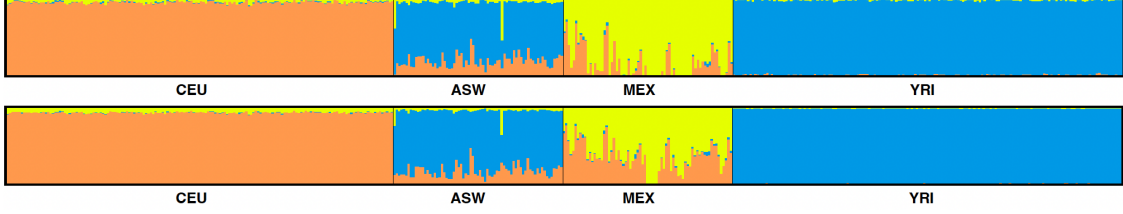


Figure 6: Membership results from STRUCTURE (upper) and the proposed method (lower) with $K = 3$ for the HapMap3 dataset. STRUCTURE takes more than 44 hours to finish while our method takes only 9 seconds. Each vertical slice represents an individual, with lengths of its three color bars proportional to the estimated $\hat{\pi}_{i,k}$.

6.2 Political Ideology: American National Election Studies

The 2022 pilot survey from the American National Election Studies (ANES) provides data about voting and public opinion after the 2022 midterm elections in the United States (publicly available at <https://electionstudies.org/>). We use a subset of $J = 145$ items after excluding followup questions, numeric response questions, and questions that directly ask about party or political ideology. A total $N = 1,511$ respondents that answered all these items is included in the analysis. The number of categories for these items ranges from 2 to 7, with the majority being 5. After flattening the polytomous response matrix, the column dimension of the data matrix becomes $J = 648$. We choose the number of extreme profiles to be $K = 3$, which is the convention in the literature for analyzing US political data (Gross and Manrique-Vallier, 2012; Wang and Erosheva, 2015). The computation time for the whole flattening and parameter estimation procedure takes 3.8 seconds.

By inspecting the estimated item response probability matrix $\hat{\Theta}$ from the proposed method, we are able to draw very intuitive interpretations on the three extreme profiles. Figure 7 demonstrates the heatmap of the estimated response probabilities corresponding to a selected subset of items with a representative category of each item. Note that these items are not necessarily binary and are indeed mostly polytomous with more than two categories. We can see that the three extreme profiles are well-differentiated by the three columns of $\hat{\Theta}$. We interpret profile $k = 1$ to be conservative in their political ideology. Respondents belonging to this extreme profile are much more likely to vote for Donald Trump, tend to care

more about illegal immigration and gun rights, and also much more in favor of the overturn of Roe v. Wade. On the contrary, profile $k = 3$ are the liberals and are much more likely to vote for Joe Biden. They also think the Democrats would do a better job handling a variety of issues such as taxes, voter fraud, Covid-19, jobs and employment, etc. Profile $k = 2$ are interpreted to be indifferent. They tend not to follow what is going on in government and public affairs, are less likely to be registered to vote.

The ANES dataset also contains self-reported party information, which we do not use for estimating the parameters. To explore how the estimated latent membership scores align with this held-out information, we present the ternary plot of $\hat{\Pi}$ color coded by the party information in the right panel in Figure 5. In this figure, each dot corresponds to a respondent, whose color represents his or her self-reported party information. We can see that Democrats are closer to the liberal profile, Republicans are closer to the conservative profile, and Independents are closer to the indifferent profile. Since we did not use the party information in the estimation procedure, this result indicates that our method extracts

Which political party would do a better job handling taxes? (Democrats)	0	0.018	1
Which political party would do a better job handling voter fraud? (Democrats)	0	0.004	1
Which political party would do a better job handling illegal immigration? (Democrats)	0	0	0.99
Which political party would do a better job handling the cost of living and rising prices? (Democrats)	0	0	0.987
Which political party would do a better job handling voting rights? (Democrats)	0	0.062	1
Which political party would do a better job handling Covid-19? (Democrats)	0	0.05	1
Which political party would do a better job handling jobs and employment? (Democrats)	0	0.036	1
How much responsibility do you think Donald Trump bears for the violence that occurred on 1/6/2021 at the US Capitol? (A great deal)	0	0.026	0.921
Which political party would do a better job handling health care? (Democrats)	0	0.143	1
Which political party would do a better job handling abortion? (Democrats)	0	0.13	1
Which political party would do a better job handling climate change? (Democrats)	0.003	0.107	1
Which political party would do a better job handling gun policy? (Democrats)	0	0.086	1
In American society, do you think that being White comes with advantages, disadvantages, or doesn't it matter? (Advantages)	0.069	0.188	1
In the 2020 presidential election, who do you believe to be the legitimate winner? (Biden)	0	0.78	1
Regarding abortion, do you usually think of yourself as pro-choice, pro-life, or something else? (Pro-choice)	0	0.56	1
How important is illegal immigration in the country today? (Extremely important)	0.948	0	0.147
How important to you is protecting the right to own guns? (Extremely important)	0.909	0	0.073
Do you favor, oppose, or neither favor nor oppose the Supreme Court's decision to overturn Roe v. Wade? (Favor)	1	0	0.033
Would you vote for Donald Trump, Joe Biden, someone else, or probably not vote? (Donald Trump)	0.916	0.076	0
How important is what's being taught in public schools in the country today? (Extremely important)	0.911	0	0.299
Does Roe v Wade being overturned make you feel outraged? (Not at all)	0.976	0.277	0
Does Roe v Wade being overturned make you feel angry? (Not at all)	0.969	0.277	0
Does Roe v Wade being overturned make you feel afraid? (Not at all)	0.956	0.314	0.041
Does Roe v Wade being overturned make you feel worried? (Not at all)	0.916	0.203	0
Do you follow what's going on in government and public affairs? (Most of the time)	0.884	0	0.863
During the past 12 months, have you posted a message or comment online about a political issue or campaign? (Yes)	0.577	0	0.7
Are you registered to vote, or not? (Yes, registered to vote at my current address)	1	0.469	0.965
	Conservative	Indifferent	Liberal

Figure 7: Heatmap of a selected subset of the estimated item response parameters with $K = 3$ for the ANES data. Each value corresponds to the estimated probability of choosing the corresponding response in the parenthesis.

meaningful and interpretable latent representations of people’s political ideology.

To demonstrate the local dependence structure, we provide a covariance matrix heatmap of the first 50 columns of the estimated error matrix $\hat{\mathbf{E}} = \mathbf{R} - \hat{\mathbf{\Pi}}\hat{\mathbf{\Theta}}^\top$ in the right panel in Figure 8. Except for the weak local dependence introduced by the flattening technique (consecutive diagonal blocks of size 3), we also observe bigger blocks of dependence. It turns out these blocks correspond to different topics of questions in the survey, such as voting participation, global emotion battery, and presidential approval. The answers to the questions within each topic are naturally dependent of each other, giving rise to the bigger blocks in the heatmap.

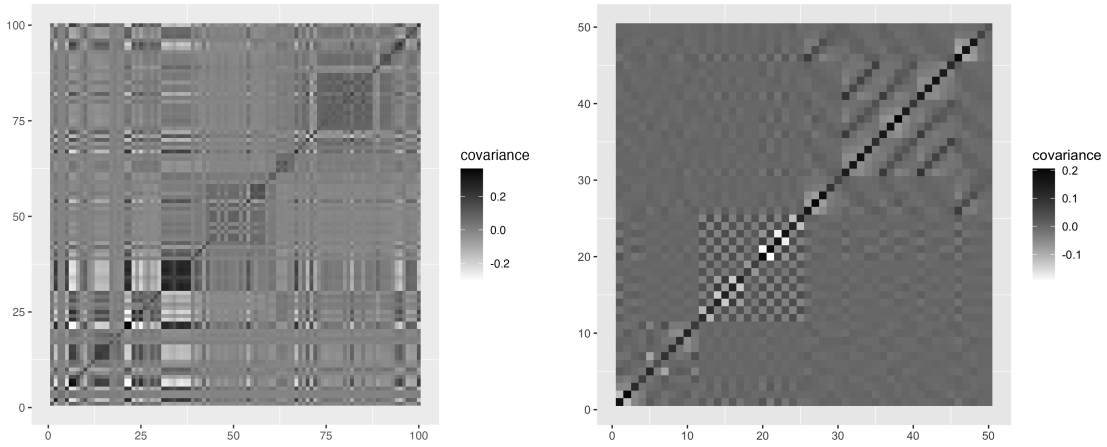


Figure 8: Heatmap of the covariance matrix of the estimated error matrix $\mathbf{R} - \hat{\mathbf{\Pi}}\hat{\mathbf{\Theta}}^\top$. The left panel corresponds to the first 100 columns of the HapMap3 data and the right panel corresponds to the first 50 columns of the ANES data.

6.3 Single-cell Genomics Data

We finally demonstrate an application to single-cell sequencing data, where there exists intrinsic local dependence between different features. The dataset contains regulatory profiles of the normal fallopian tube at single-cell resolution with $N = 259$ cells and $J = 22,431$ genes (Deng et al., 2014, publicly available at <https://cellxgene.cziscience.com/collections/d36ca85c-3e8b-444c-ba3e-a645040c6185>). This dataset also provides labels of 10 pre-implantation development phase categories for each cell.

The gene expression counts in our dataset range from 0 to 693,974, and we assume

a Poisson model. In single-cell sequencing data, the phenomenon of local dependence is commonly referred to as gene co-expression (Van Der Wijst et al., 2018). We choose $K = 6$ according to the analysis on the same dataset conducted in Dey et al. (2017). Figure 9 presents our estimated membership scores $\hat{\Pi}$, with each vertical slice corresponding to a cell. From left (zygote) to right (late blast) are the 10 cell development phases. We can see that cells belonging to the same phase have relatively similar mixed membership allocations, and that the early stage cells are less mixed compared to cells at later stages. Furthermore, we observe a continuous evolution in the membership scores as the cells progress through the developmental stages. These findings align with those reported in Dey et al. (2017). Additionally, it takes 0.3 second to run the proposed method on the count matrix, while the computation time is reported to be 12 minutes in Dey et al. (2017), where they use R package `maptp` (Taddy, 2012) to fit the GoM model.

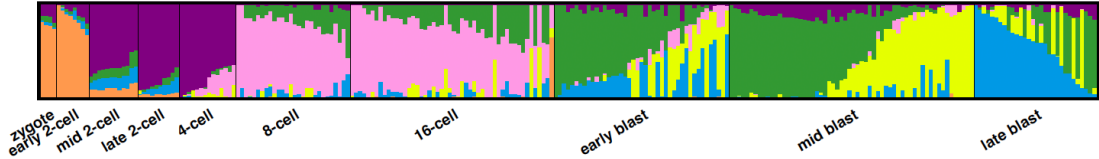


Figure 9: Generalized-GoM estimated memberships for the single cell data with $K = 6$. Each color represents an extreme latent profile, and each vertical slice represents a cell. The cells are ordered by their preimplantation development phase, and within each phase sorted by the membership score of the dominating profile in this phase.

7 Discussion

This paper introduces an efficient spectral estimation method for generalized GoM models with local dependence. For GoM models for high-dimensional polytomous data, we flatten the three-way quasi-tensor of data into a fat binary matrix and exploit its singular subspace geometry to estimate parameters. We establish fine-grained entrywise error bounds for the membership scores and item parameters. We also develop a novel $\ell_{2,\infty}$ singular subspace perturbation theory for arbitrary locally dependent and flexibly distributed noise, which is of independent interest. Our method has a significant computational advantage over popular

Bayesian MCMC methods and outputs competitively accurate parameter estimates. Even if one ultimately desires to perform Bayesian posterior inference, we recommend initializing the MCMC algorithm using our spectral method. This sufficiently accurate initialization can help speed up the convergence of the MCMC chain to its stationary distribution.

There are several future directions worth exploring. First, Algorithm 1 requires the input of K , which is often unknown in practice. There are various tools to determine the choice of K . The simple yet widely-used approach is to inspect the scree plot of the singular values and find the “elbow point” (Cattell, 1966). Other methods include gap statistic (Tibshirani et al., 2001), parallel analysis (Hayton et al., 2004), eigen selection (Han et al., 2023). It is of special interest to apply the proposed method on an estimated K and still obtain theoretical guarantees, similar to the work in Fan et al. (2022) and Jin et al. (2023). Second, missing data can appear in real-world applications especially with big data scenarios. Matrix completion using spectral methods has been studied (Chen and Li, 2022; Cai et al., 2021) and could be potentially extended to our setting. Third, to further increase the modeling power of the GoM model, a potential extension is to incorporate the degree-correction parameter similar to the degree-corrected mixed membership model for networks (Ke and Jin, 2023; Jin et al., 2024). Then one can perform certain normalizations in the spectral domain to alleviate the effect of degree heterogeneity before estimating other parameters. Finally, it would be interesting to develop statistical inference results for locally dependent generalized-GoM models, similar to the considerations in Fan et al. (2022) and Du and Tang (2023).

Supplementary Material

The Supplementary Material contains all technical proofs of the theoretical results and additional computation details.

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